

offset the impact of the increase in foreign exchange reserves on the money supply.

5.2.3 Equilibrium in Open Economy IS-LM Model

As in case of a closed economy, macroeconomic equilibrium occurs at the intersection of the IS and LM curves as shown in Fig. 5.1.

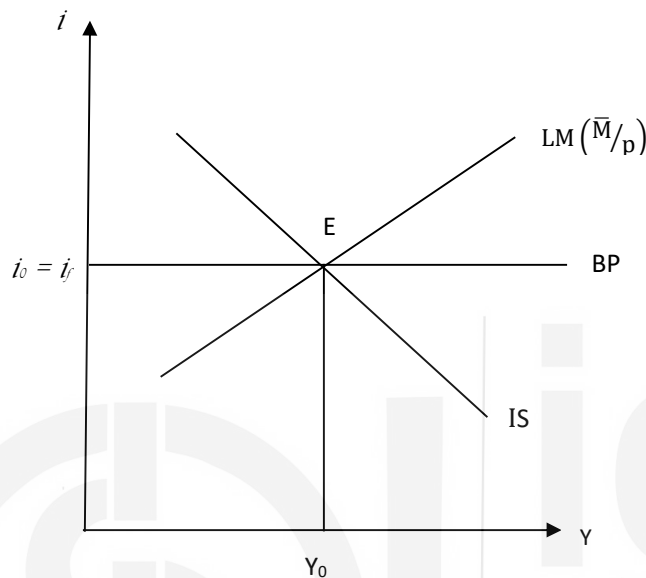


Fig. 5.1: “Macroeconomic Equilibrium in Open Economy IS-LM Model

Note that at the equilibrium E, the rate of interest, $i_0 = i_f$, owing to the assumption of perfect capital mobility. The point E represents macroeconomic equilibrium in the sense that there is both internal balance (goods and money market equilibrium) and external balance (BOP equilibrium).

The horizontal line BP that passes through the equilibrium rate of interest represents BOP equilibrium, in the sense that it represents interest rate – income combinations for which the current account surplus (deficit) is exactly offset by a capital account deficit (surplus). Note that here NX represents the current account balance.

With perfect capital mobility, the BP curve is horizontal, as the slightest deviation of i from i_f , would create BOP disequilibrium and trigger either infinite capital inflows or outflows. If $i > i_f$, there will be a surge in capital inflows and a BOP surplus, since inflows would be higher than required for BOP balance, at any given level of income. Whereas, or any $i < i_f$, there will be unlimited capital outflows and a BOP deficit. That is, for any Y (say Y_0), there is BOP surplus at all points above BP and a BOP deficit at all points below BP.

With imperfect capital mobility the BP curve would be upward sloping. As Y increases, M would rise, NX would fall and the current account deficit would widen. To maintain BOP balance therefore, i would have to increase to attract capital inflows and create a matching capital account surplus. With imperfect

capital mobility, which occurs when there are restrictions on cross-border capital flows or when domestic and foreign bonds are imperfect substitutes, there can be a difference between i and i_f . However, here we will only consider the case of perfect capital mobility.

5.3 MONETARY AND FISCAL POLICIES UNDER FLEXIBLE EXCHANGE RATE

When a country has a flexible exchange rate regime, its exchange rate is determined by demand and supply in the foreign exchange market. We consider the case of a *pure float*, where there is no official intervention in currency markets and hence no role of foreign exchange reserves, with BOP equilibrium achieved via exchange rate fluctuations.

5.3.1 Fiscal Policy under Flexible Exchange Rates

Consider the open economy IS-LM model with perfect capital mobility and suppose the government follows expansionary fiscal policy in this set up as in Fig. 5.2.

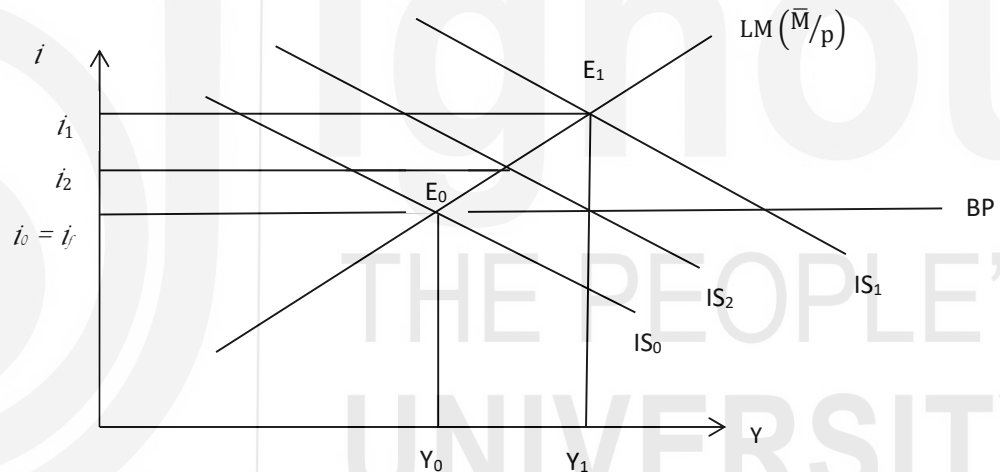


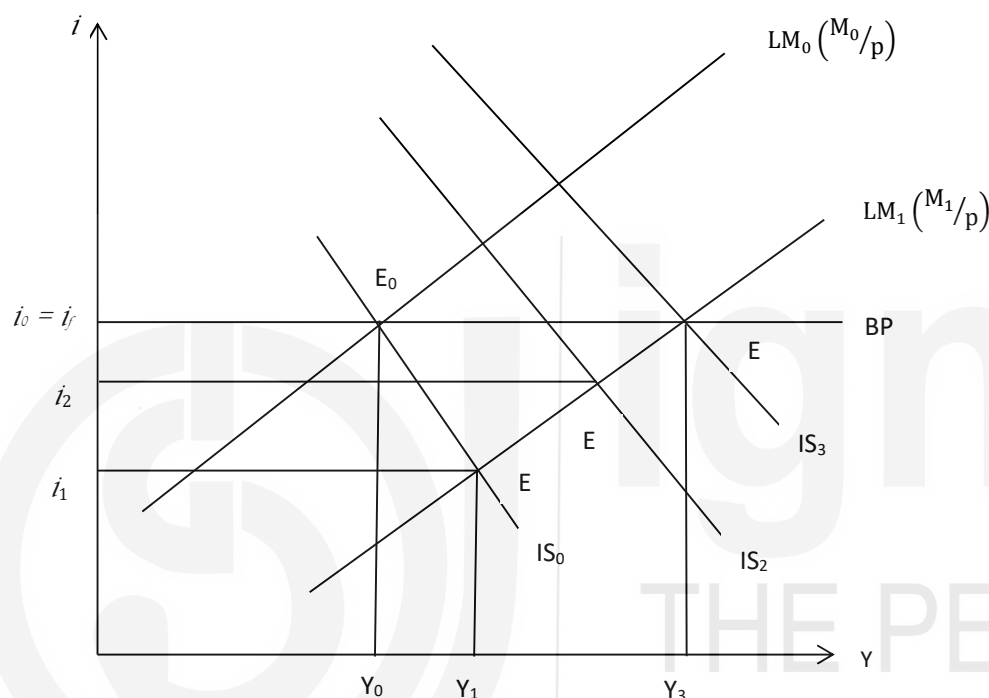
Fig. 5.2: Expansionary Fiscal Policy under Flexible Exchange Rates

Initially the economy is at E_1 (see Fig. 5.2), where goods and money markets are in equilibrium and there is BOP equilibrium as well, with $i_0 = i_f$. With fiscal expansion (or fall in tax rate t), there is an increase in G . The IS curve shifts from IS_0 to IS_1 and the interest rises from i_0 to i_1 as the economy moves to the new equilibrium E_1 . However, with i_f remaining the same, the rise in domestic interest rate makes domestic bonds relatively more attractive and there is a surge in capital inflows, that leads to a BOP surplus (excess supply of foreign exchange), causing a currency appreciation (e falls). With given prices (implying prices remaining constant), a nominal appreciation leads to a real appreciation (r falls), reducing X , raising M and worsening the trade balance (NX falls). As a result aggregate demand falls and the IS curve shifts to the left. It shifts back all the way to IS_0 , where once again $i = i_0 = i_f$ and output is back to Y_0 . Note that at any other IS curve, e.g., IS_2 , the domestic interest rate is still above i_f ($i_2 > i_f$), so e , r

and NX would fall and IS would shift further to the left. Therefore, we see that with flexible exchange rates, fiscal expansion is ineffective. It fails to raise output (the increase in output from Y_0 to Y_1 is only temporary), the trade balance worsens because of appreciation in the exchange rate (NX is lower) and the fiscal position deteriorates (G has increased or tax rate t has fallen).

5.3.2 Monetary Policy under Flexible Exchange Rates

We now examine the impact of expansionary monetary policy under a flexible exchange rate regime.



[Fig. 5.3: Expansionary Monetary Policy under Flexible Exchange Rates]

In Fig. 5.3 an increase in nominal money supply (from M_0 to M_1) shifts the LM curve outward from LM_0 to LM_1 and the economy moves from E_0 to E_1 , as domestic interest rate falls from i_0 to i_1 . At i_1 there is internal balance (output and money markets are in equilibrium) but there is external imbalance (BOP disequilibrium) that triggers capital flows. Since $i_1 < i_f$, this triggers a capital outflow, leading to a BOP deficit (excess demand for dollars) and causing a depreciation of the nominal (e rises) and real exchange rates (r rises). As a result domestic goods become relatively cheaper, X rises, M falls and NX increases, and with higher external demand, the IS curve shifts to the right. This process continues till the IS curve shifts all the way to IS_3 and the economy is at E_3 , where once again $i = i_f$ and both internal and external balance are restored. At any other point (e.g., E_2 on IS_2 , $i_2 < i_f$, so e and r would rise, NX would fall, moving the IS curve further to the right).

This shows, that in complete contrast to fiscal policy, monetary expansion is an extremely effective policy tool in an open economy with flexible exchange rates